Meeting 03.07

Big Picture

The Art of Making Problems Simple: A Theory of Intelligence

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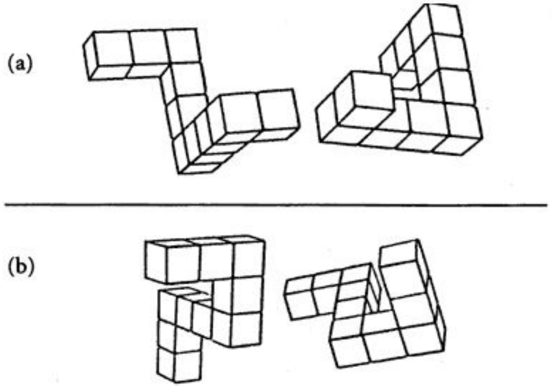
Abstract

Human intelligence is flexible and general, yet its underlying structure remains poorly understood. We propose that intelligence arises from the ability to construct and revise representational normal forms, that is, coordinate systems in which prediction, comparison, and inference become simple. This capacity rests on four operations: identifying lawful generators of variation, quotienting task-irrelevant differences, constructing measurable axes for ordered reasoning, and selecting representations that locally flatten the space of possibilities.

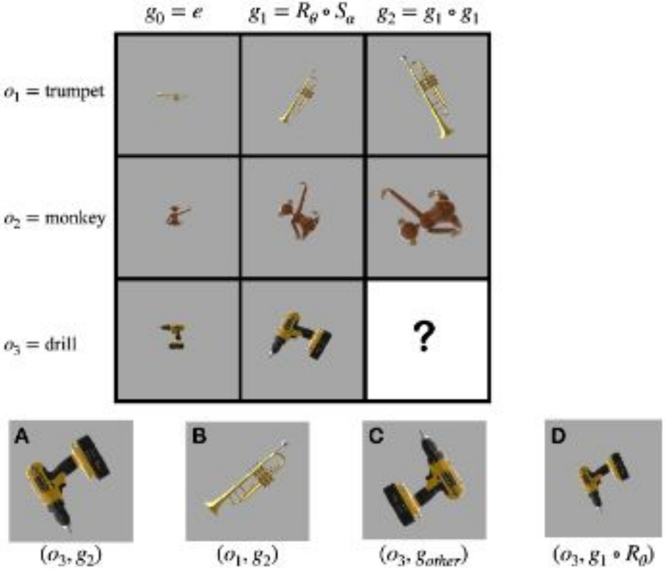
"intelligence is ability to construct and revise representational normal forms"

(1) discovering lawful generators of variations in the world
(2) quotienting task-irrelevant distinctions
(3) …

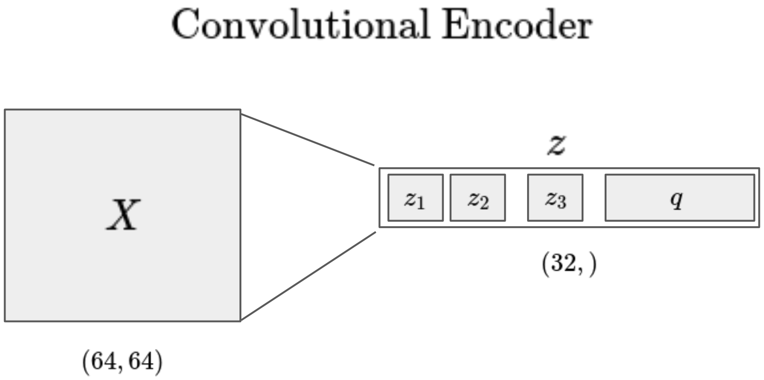
Big Picture



Mental Rotation Test—Are these two figures the same except for their orientation?



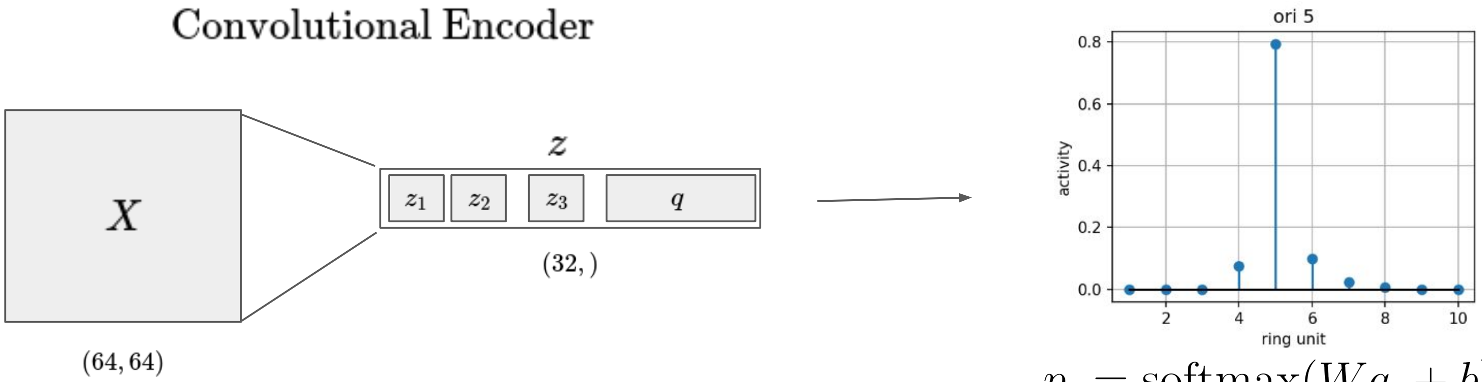
Gap & Goal



Project's goal:

extend to 3D rotations

Approach

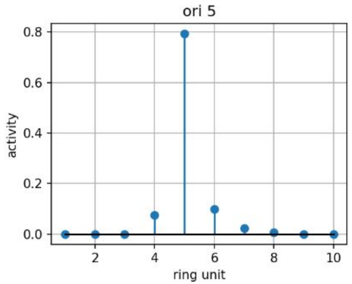


$$p_t = \mathrm{softmax}(W q_t + b)$$

Approaches

Route A (Prototype orbit approach)

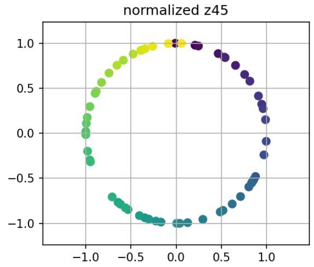
- orientation as a **bump distribution on a discrete ring**
- output: $p_t \in \mathbb{R}^K$
- orientation as **activity over ring units**
- decoder: similarity to orientation prototypes



$$p_t = \mathrm{softmax}(Wq_t + b)$$

Route B (Continuous vector-feature approach)

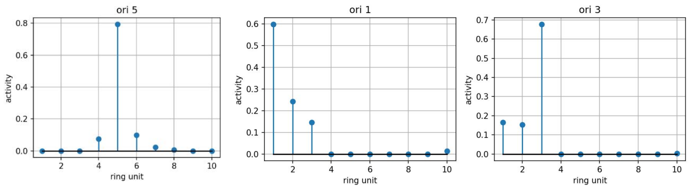
- orientation as a **continuous 2D circular coordinate**
- output: $(u_t, v_t) \in \mathbb{R}^2$
- orientation as **position on a learned circle**
- decoder: learned neural mapping

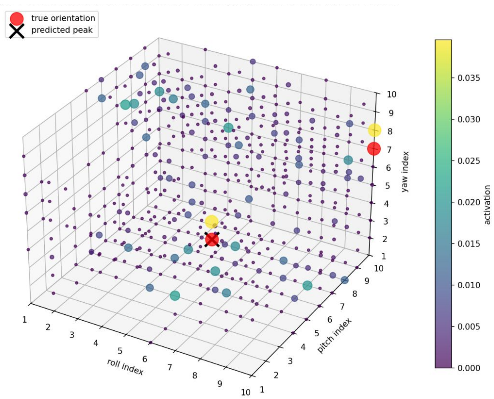


Obtain the **3d bump distribution** by training a **neural network** instead of looking at similarities to prototypes

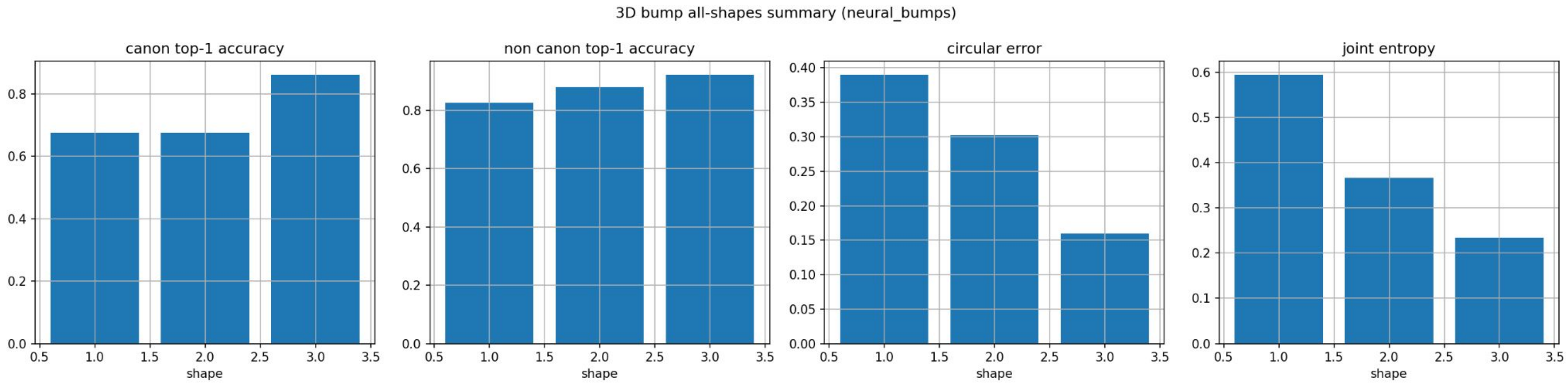
$$p_{roll} = \mathrm{softmax}(Wq + b) \ \epsilon \ \mathbb{R}$$
$$p_{pitch} = \mathrm{softmax}(Wq + b) \ \epsilon \ \mathbb{R}$$
$$p_{yaw} = \mathrm{softmax}(Wq + b) \ \epsilon \ \mathbb{R}$$

$$p_{joint} = \mathrm{softmax}(\mathrm{net}(q)) \ \epsilon \ \mathbb{R}^3$$





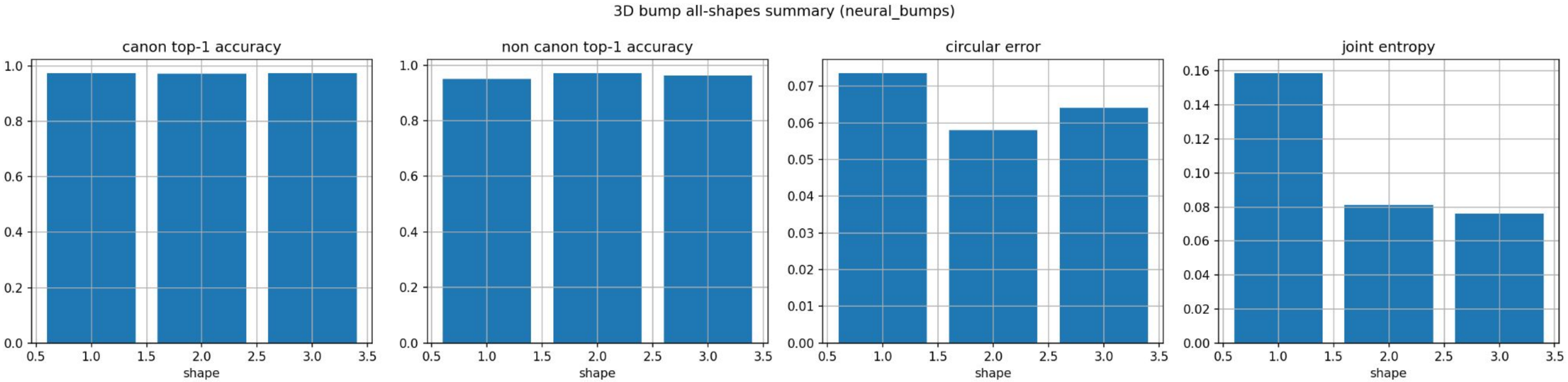
Results



3D bump all-shapes summary (neural_bumps)
canon top-1 accuracy
non canon top-1 accuracy
circular error
joint entropy
shape
shape
shape
shape

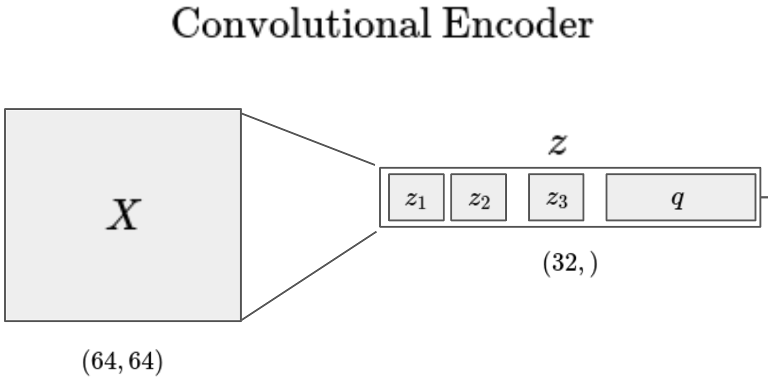
Next steps

Enlarging the dataset

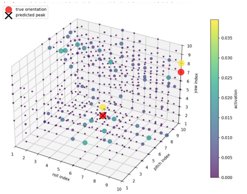


With the encoder trained on 6 shapes and only in-plane rotation

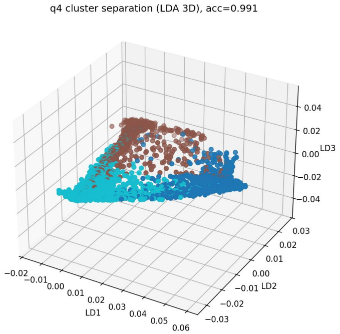
Questions



discovering bump distribution



classifying shapes



Questions

In this approach, orientation decoding and shape classification are performed in parallel, not successively. How is this compatible with the general idea of this project?

From Angela’s project description:

“The pipeline demonstrates that transformation generators can be discovered and represented explicitly within a learned representation. By constructing explicit coordinates for nuisance transformations and factoring them from object identity, the resulting quotient representation supports substantially simpler downstream inference.”

Questions

Is it justifiable to divert from the original prototype-orbit idea to a neural network?

From Angela's project description:

“**7. Key hypothesis**

For transformations with **occlusion and appearance discontinuities**, as as generalization, the **prototype-based orbit approach may perform better**, because it:

- models the orbit **locally**
- does not require a globally smooth appearance manifold
- supports **compositional chaining of local transitions**

This would support the broader ToI claim that **representational innovation often begins with local structure rather than globally smooth embeddings**.”